

DATA130005 统计学基础 I：数理统计 Assignment 1

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Problem1 习题 E1.1

表 1.1-2

姓 名	年龄	身高 (厘米)	性别	体重 (千克)	姓 名	年龄	身高 (厘米)	性别	体重 (千克)
LAWRENCE	17	172	男	78.1	JOE	13	154	男	47.7
JEFFERY	14	169	男	51.3	MARY	15	152	女	41.8
EDWARD	14	167	男	50.8	LINDA	17	152	女	52.7
PHILLIP	16	167	男	58.1	MARK	15	152	男	47.2
KIRK	17	167	男	60.8	PATTY	14	152	女	38.6
ROBERT	15	164	男	58.1	ELIZABET	14	152	女	41.3
JACLYN	12	162	女	65.8	JUDY	14	149	女	36.8
DANNY	15	162	男	48.1	LOUISE	12	149	女	55.8
CLAY	15	162	男	47.7	ALICE	13	149	女	48.6
HENRY	14	159	男	54	JAMES	12	149	男	58.1
LESLIE	14	159	女	64.5	MARIAN	16	147	女	52.2
JOHN	13	159	男	44.5	TIM	12	147	男	38.1
WILLIAM	15	159	男	50.4	BARBARA	13	147	女	50.8
MARTHA	16	159	女	50.8	DAVID	13	145	男	35.9
LEWIS	14	157	男	41.8	KATIE	12	145	女	43.1
AMY	15	157	女	50.8	MICHAEL	13	142	男	43.1
ALFRED	14	157	男	44.9	SUSAN	13	137	女	30.4
CHRIS	14	157	男	44.9	JANE	12	135	女	33.6
FREDRICK	14	154	男	42.2	LILLIE	12	127	女	29.1
CAROL	14	154	女	38.1	ROBERT	12	125	男	35.9

我们用以下 R 代码生成包含表1.1-2数据的数据框：

```
1 students <- data.frame(  
2   name = c("LAWRENCE", "JEFFERY", "EDWARD", "PHILLIP", "KIRK",  
3           "ROBERT", "JACLYN", "DANNY", "CLAY", "HENRY",  
4           "LESLIE", "JOHN", "WILLIAM", "MARTHA", "LEWIS",  
5           "AMY", "ALFRED", "CHRIS", "FREDRICK", "CAROL",  
6           "JOE", "MARY", "LINDA", "MARK", "PATTY",  
7           "ELIZABET", "JUDY", "LOUISE", "ALICE", "JAMES",  
8           "MARIAN", "TIM", "BARBARA", "DAVID", "KATIE",  
9           "MICHAEL", "SUSAN", "JANE", "LILLIE", "ROBERT"),  
10  age = c(17, 14, 16, 16, 17, 15, 12, 15, 15, 14,  
11          14, 13, 15, 16, 14, 15, 14, 14, 14, 14,  
12          13, 15, 17, 15, 14, 14, 14, 12, 13, 12,  
13          16, 12, 13, 13, 12, 13, 13, 12, 12, 12),  
14  height = c(172, 169, 167, 167, 167, 164, 162, 162, 162, 159,  
15             159, 159, 159, 159, 157, 157, 157, 157, 154, 154,  
16             154, 152, 152, 152, 152, 152, 152, 149, 149, 149,  
17             147, 147, 147, 145, 145, 142, 137, 135, 127, 125),  
18  gender = c("M", "M", "M", "M", "M", "M", "F", "M", "M", "M",  
19             "F", "M", "M", "F", "M", "F", "M", "M", "M", "F",  
20             "M", "F", "F", "M", "F", "F", "F", "F", "F", "M",
```

```

21     "F", "M", "F", "M", "F", "M", "F", "F", "F", "M"),
22     weight = c(78.1, 51.3, 50.8, 58.1, 60.8, 58.1, 65.8, 48.1, 47.7, 54,
23               64.5, 44.5, 50.4, 50.8, 41.8, 50.8, 44.9, 44.9, 42.2, 38.1,
24               47.7, 41.8, 52.7, 47.2, 38.6, 41.3, 36.8, 55.8, 48.6, 58.1,
25               52.2, 38.1, 50.8, 35.9, 43.1, 43.1, 30.4, 33.6, 29.1, 35.9)
26 )

```

(1) 按学生姓名的第一个字母分组，统计不同组包含学生的频数.

R 代码:

```

1  # Extract the first letter of each name
2  first_letters <- substr(students$name, 1, 1)
3
4  # Calculate the frequency of each first letter
5  letter_frequency <- table(first_letters)
6
7  # Print the frequency table
8  print(letter_frequency)

```

运行结果:

```

1  first_letters
2  A B C D E F H J K L M P R S T W
3  3 1 3 2 2 1 1 7 2 6 5 2 2 1 1 1

```

(2) 计算体重数据的各个汇总统计量 (均值、方差、标准差、中位数、极差)

R 代码:

```

1  # mean
2  mean_weight <- mean(students$weight)
3
4  # variance
5  variance_weight <- var(students$weight)
6
7  # standard deviation
8  sd_weight <- sd(students$weight)
9
10 # median
11 median_weight <- median(students$weight)
12
13 # range
14 range_weight <- range(students$weight)
15 range_value <- diff(range_weight)
16
17 # print statistics
18 cat("mean:", mean_weight, "\n")
19 cat("variance:", variance_weight, "\n")
20 cat("standard deviation:", sd_weight, "\n")

```

```
21 cat("median:", median_weight, "\n")
22 cat("minimum:", range_weight[1], "\n")
23 cat("maximum:", range_weight[2], "\n")
24 cat("range:", range_value, "\n")
```

运行结果:

```
1 mean: 47.6625
2 variance: 101.4886
3 standard deviation: 10.07415
4 median: 47.7
5 minimum: 29.1
6 maximum: 78.1
7 range: 49
```

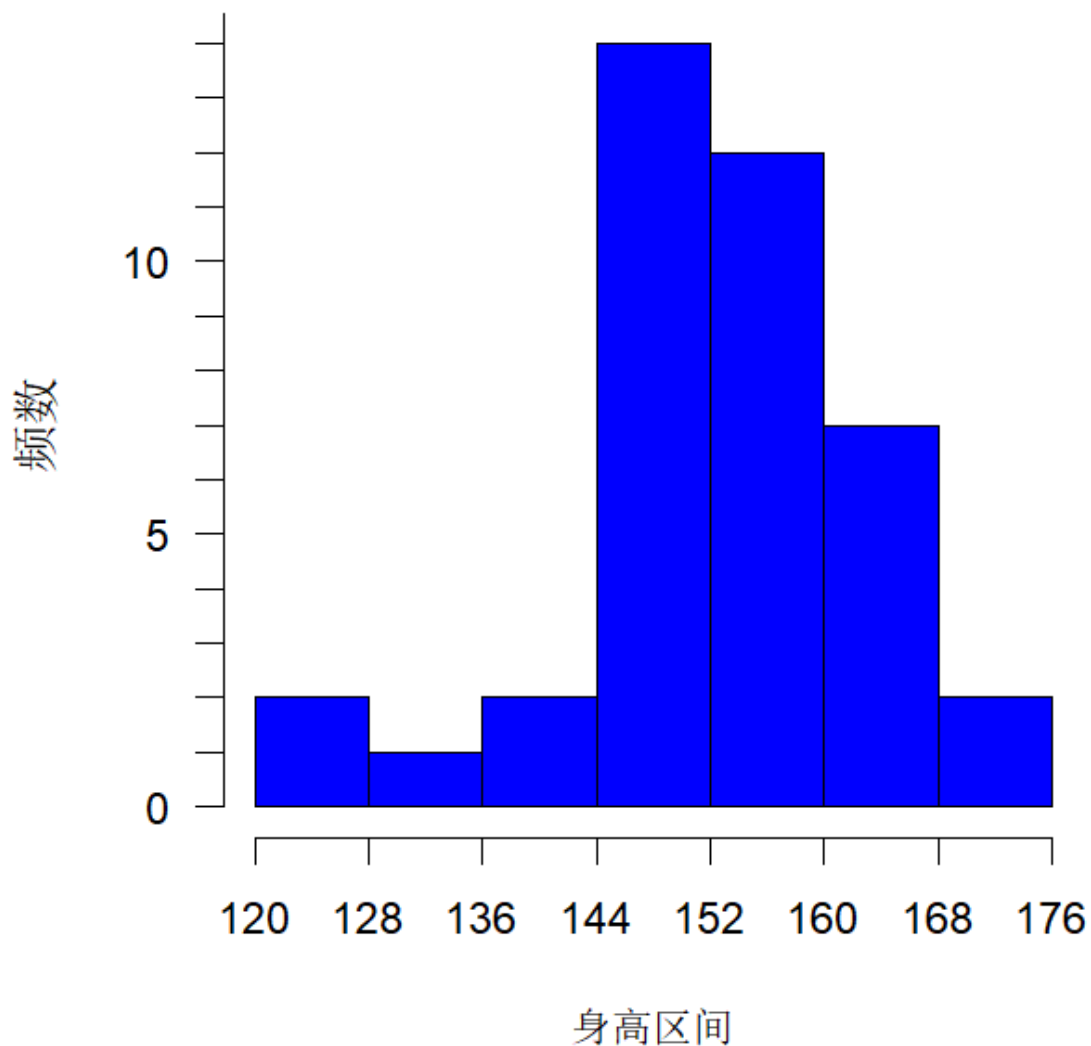
(3) 将身高数据按 $(120, 128]$, $(128, 136]$, $\dots$, $(168, 176]$ 分组绘制直方图

R 代码:

```
1 # Define the boundaries of groups
2 breaks <- seq(120, 176, by=8)
3
4 # Create a histogram
5 hist(students$height, breaks=breaks, main="身高分布直方图",
6      xlab="身高区间", ylab="频数",
7      col="blue", border="black",
8      right=TRUE, # which means the interval type is (]
9      xaxt='n', yaxt='n' # disable the default axis
10     )
11
12 # User-defined axis
13 # horizontal axis
14 axis(1, at=seq(120, 176, by=8), labels=seq(120, 176, by=8), las=1)
15 # vertical axis
16 axis(2, at=seq(0, 20, by=1), las=1,
17      labels=ifelse(seq(0, 20, by=1) %in% seq(0, 20, by=5), seq(0, 20, by=1),
18      ""))
18 )
```

运行结果：

身高分布直方图



Problem 2 习题 E1.5

设样本量为 10 的样本的观测值为 0.5, 0.7, 0.2, 0.7, 0.4, 2.5, 1.5, -0.2, -0.5, 0.1
试绘制其经验分布函数的图形.

R 代码:

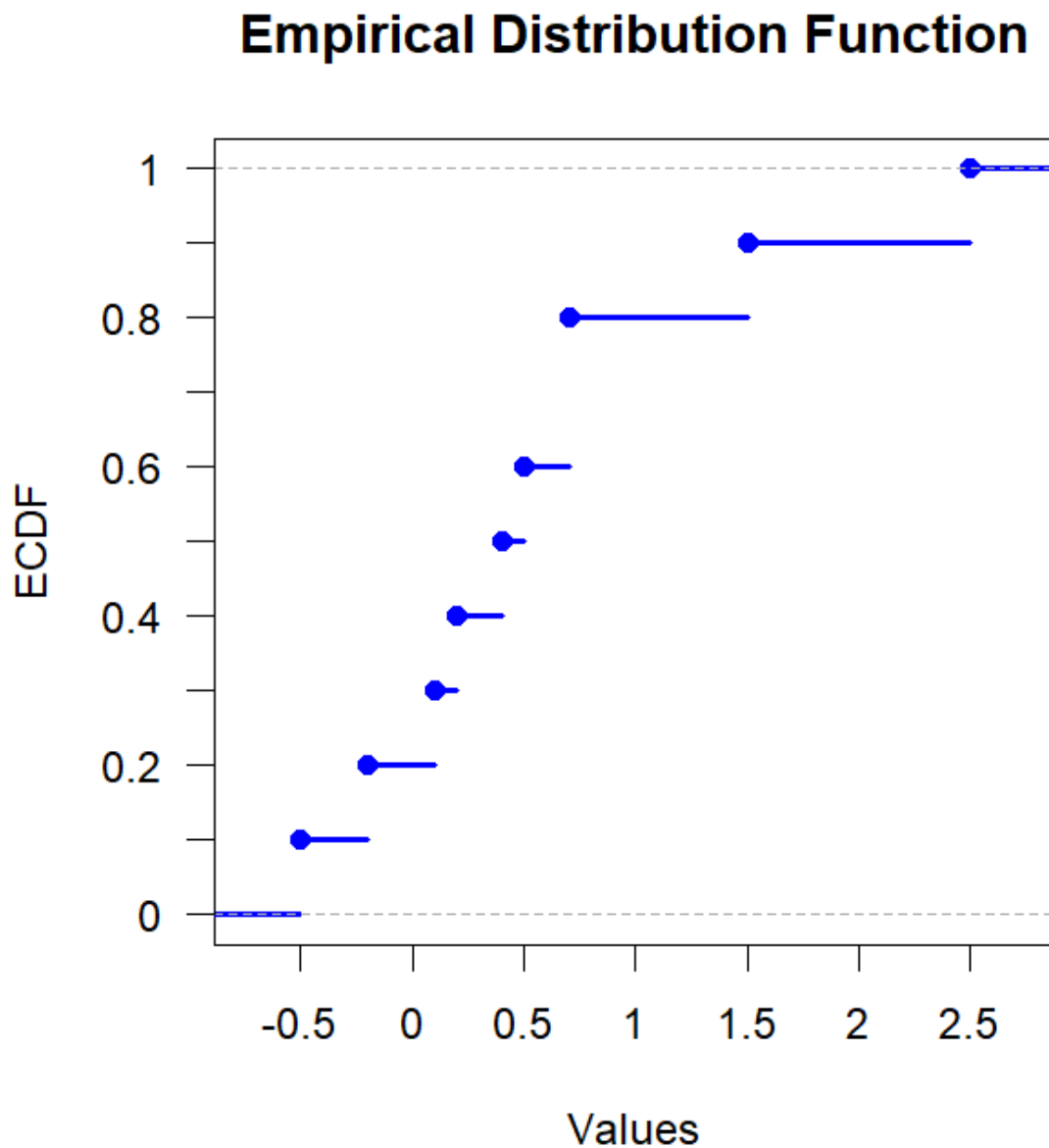
```
1 # sample data
2 sample_data <- c(0.5, 0.7, 0.2, 0.7, 0.4, 2.5, 1.5, -0.2, -0.5, 0.1)
3
4 # calculate Empirical Distribution Function
5 sample_ecdf <- ecdf(sample_data)
6
7 # create Empirical Distribution Function
8 plot(sample_ecdf, main="Empirical Distribution Function",
9       xlab="Values", ylab="ECDF", col="blue", lwd=2,
10      xaxt='n', yaxt='n' # disable the default axis
11 )
12
```

```

13 # User-defined axis
14 # horizontal axis
15 axis(1, at=seq(-1, 3, by=0.5), labels=seq(-1, 3, by=0.5), las=1)
16 # vertical axis
17 axis(2, at=seq(0, 1, by=0.1), las=1,
18       labels=ifelse(seq(0, 1, by=0.1) %in% seq(0, 1, by=0.2), seq(0, 1,
19       by=0.1), ""))
19 )

```

运行结果:



Problem 3 习题 E1.6

任意给定常数 $a \neq 0$ 和 b , $y_i = ax_i + b$ ($i = 1, \dots, n$)

(1) 证明: $\begin{cases} x_1, \dots, x_n \\ y_1, \dots, y_n \end{cases}$ 的样本均值、方差之间有 $\begin{cases} \bar{y} = a\bar{x} + b \\ s_y^2 = a^2 s_x^2 \end{cases}$ 成立

- ① 证明 $\bar{y} = a\bar{x} + b$:

$$\begin{aligned}\bar{y} &= \frac{1}{n} \sum_{i=1}^n y_i \\ &= \frac{1}{n} \sum_{i=1}^n ax_i + b \\ &= a\bar{x} + b\end{aligned}$$

- ② 证明 $s_y^2 = a^2 s_x^2$:

$$\begin{aligned}s_y^2 &= \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (ax_i + b - a\bar{x} - b)^2 \\ &= \frac{1}{n-1} a^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= a^2 s_x^2\end{aligned}$$

(2) 根据 (1) 的结果, 使用适当的线性变换, 求下列一组数据的样本均值、方差:

480, 550, 510, 590, 510, 610, 490, 600, 580

取 $a = 0.1, b = -48$

得到新数据: 0, 7, 3, 11, 3, 13, 1, 12, 10

可以计算其样本均值、方差为 $\begin{cases} \bar{y} = \frac{60}{9} = 6\frac{2}{3} \\ s_y^2 = \frac{202}{8} = 25.25 \end{cases}$

于是有 $\begin{cases} \bar{x} = \frac{1}{a}(\bar{y} - b) = 10(6\frac{2}{3} + 48) \approx 546.6667 \\ s_x^2 = \frac{1}{a^2}s_y^2 = 2525 \end{cases}$

Problem 4 习题 E1.8

若样本观测值 $x_1, \dots, x_m$ 的频数分别为 $n_1, \dots, n_m$,

试写出计算样本均值 $\bar{X}$ 和样本方差 S_n^2 的公式 (其中 $n = n_1 + \dots + n_m$)

解:

- 样本均值:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^m n_i x_i$$

- 样本方差：

$$\begin{aligned}
 S_n^2 &= \frac{1}{n-1} \sum_{i=1}^m n_i (x_i - \bar{X})^2 \\
 &= \frac{1}{n-1} \sum_{i=1}^m n_i (x_i^2 - 2\bar{X}x_i + \bar{X}^2) \\
 &= \frac{1}{n-1} \left\{ \left(\sum_{i=1}^m n_i x_i^2 \right) - 2\bar{X} \sum_{i=1}^m n_i x_i + n\bar{X}^2 \right\} \\
 &= \frac{1}{n-1} \left\{ \left(\sum_{i=1}^m n_i x_i^2 \right) - n\bar{X}^2 \right\}
 \end{aligned}$$

Problem 5

对以下金融铱的升华热数据：

```

1 # sample data
2 sublimation_heat_data <- c(
3   136.6, 145.2, 151.5, 162.7, 159.1,
4   159.8, 160.8, 173.9, 160.1, 160.4,
5   161.1, 160.6, 160.2, 159.5, 160.3,
6   159.2, 159.3, 159.6, 160.0, 160.2,
7   160.1, 160.0, 159.7, 159.5, 159.5,
8   159.6, 159.5
9 )

```

(1) 制作直方图

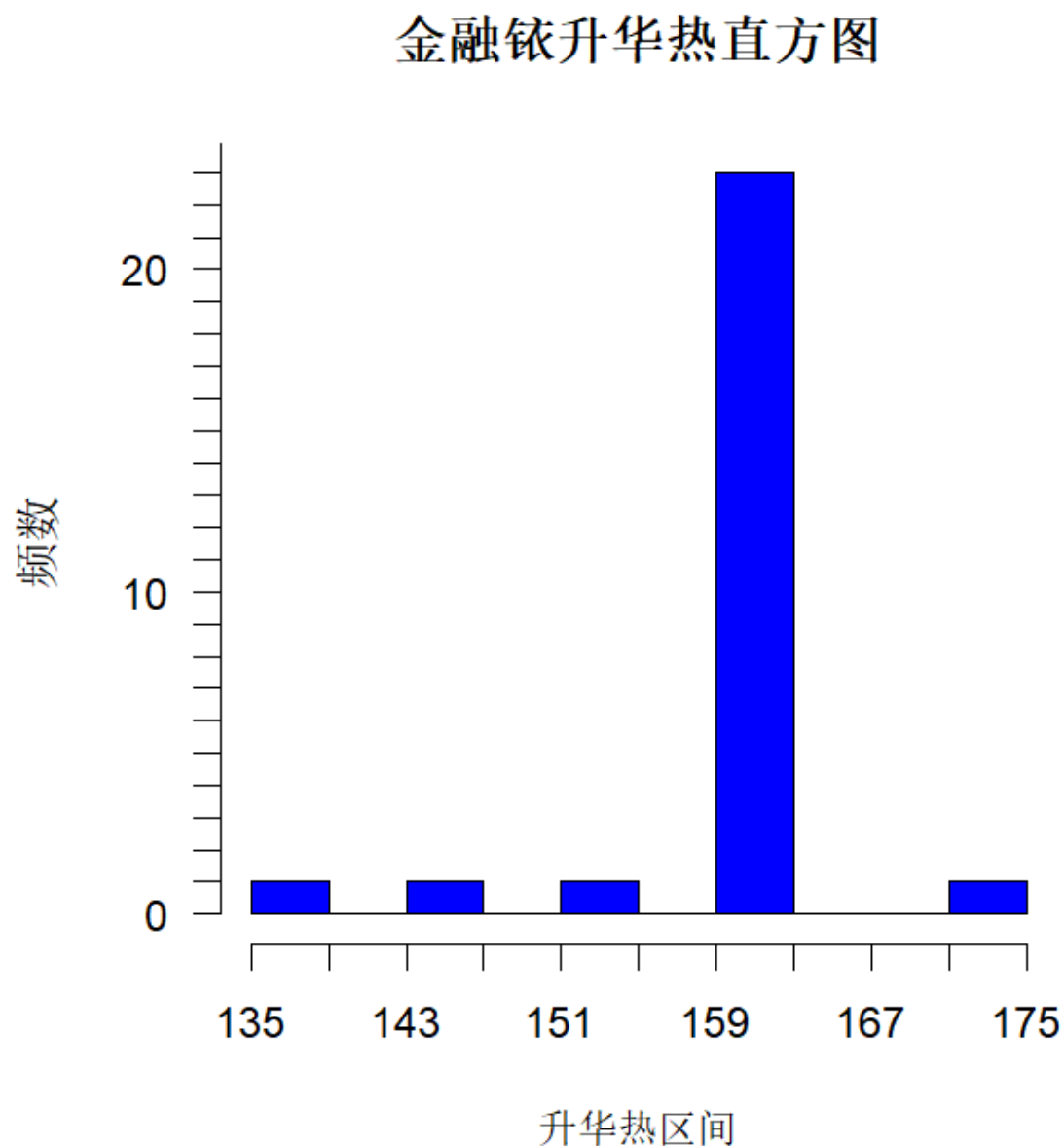
R 代码：

```

1 # Define the boundaries of groups
2 breaks <- seq(135, 175, by=4)
3
4 # Create a histogram
5 hist(sublimation_heat_data, breaks=breaks,
6     main="金融铱升华热直方图",
7     xlab="升华热区间", ylab="频数",
8     col="blue", border="black",
9     xaxt='n', yaxt='n' # disable the default axis
10 )
11
12 # User-defined axis
13 # horizontal axis
14 axis(1, at=seq(135, 175, by=4), labels=seq(135, 175, by=4), las=1)
15 # vertical axis
16 axis(2, at=seq(0, 30, by=1), las=1,
17     labels=ifelse(seq(0, 30, by=1) %in% seq(0, 30, by=10), seq(0, 30,
18     by=1), ""))
19 )

```

运行结果：



(2) 制作经验分布图

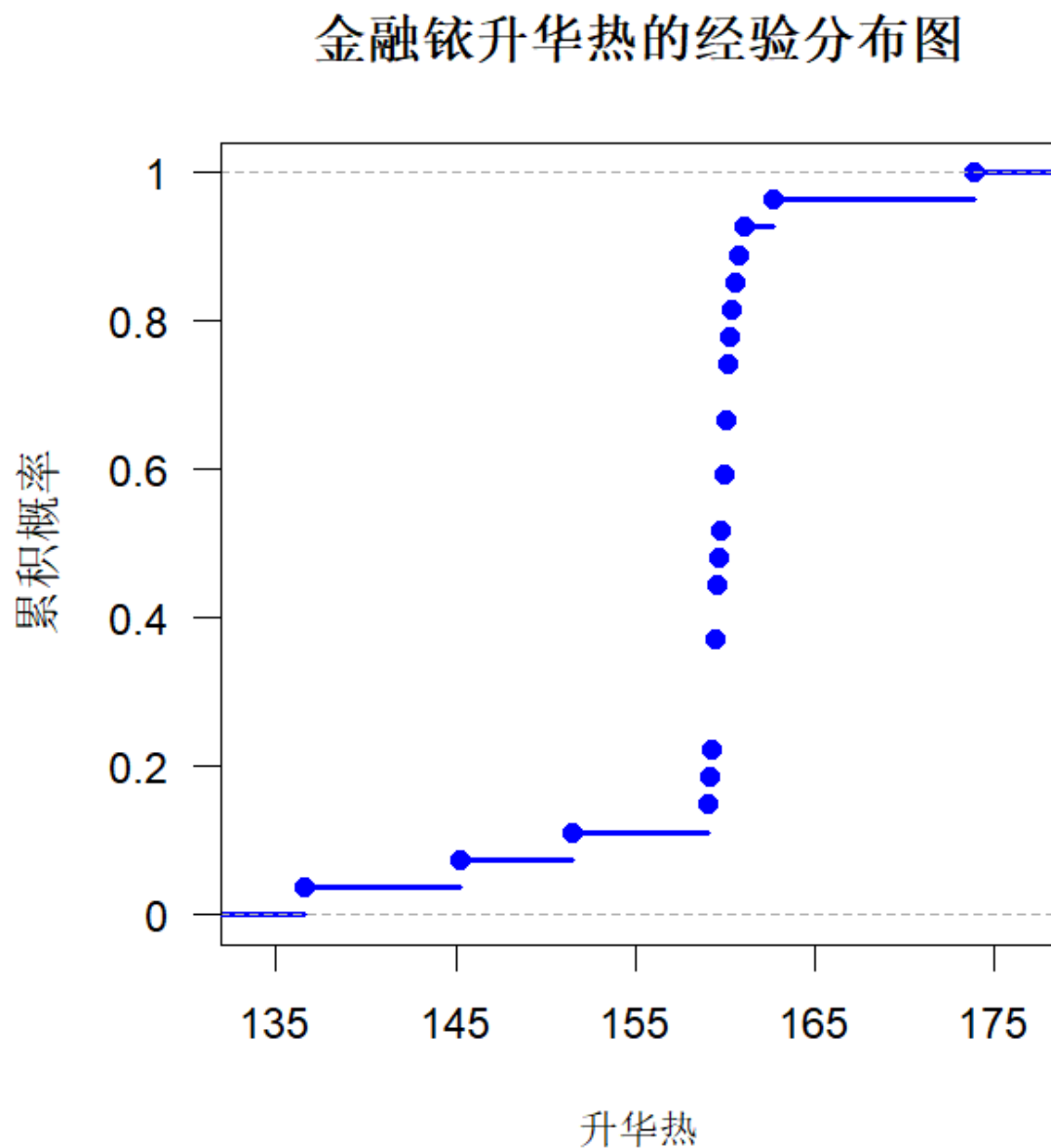
R 代码：

```
1 # calculate Empirical Distribution Function
2 edf <- ecdf(sublimation_heat_data)
3
4 # create Empirical Distribution Function
5 plot(edf, main="金融铱升华热的经验分布图",
6       xlab="升华热", ylab="累积概率",
7       col="blue", lwd=2,
8       xaxt='n', yaxt='n' # disable the default axis
9     )
10
11 # User-defined axis
12 # horizontal axis
13 axis(1, at=seq(135, 175, by=10), labels=seq(135, 175, by=10), las=1)
14 # vertical axis
```



```
15 axis(2, at=seq(0, 1, by=0.2), labels=seq(0, 1, by=0.2), las=1)
```

运行结果：



(3) 给出其在 0.90, 0.75, 0.25, 0.05, 0.01 的分位数

R 代码：

```
1 # 指定的概率水平
2 prob_levels <- c(0.90, 0.75, 0.25, 0.05, 0.01)
3
4 # 计算并打印分位数
5 quantiles <- quantile(sublimation_heat_data, probs = prob_levels)
6 print(quantiles)
```

运算结果：

1	90%	75%	25%	5%	1%
2	160.920	160.250	159.500	147.090	138.836

(4) 计算其样本均值、方差、标准差、偏态系数和峰态系数

R 代码:

```
1 library(e1071)
2 mean_value <- mean(sublimation_heat_data)
3 variance_value <- var(sublimation_heat_data)
4 std_deviation <- sd(sublimation_heat_data)
5 skewness_value <- skewness(sublimation_heat_data)
6 kurtosis_value <- kurtosis(sublimation_heat_data)
7
8 cat("mean: ", mean_value, "\n")
9 cat("variance: ", variance_value, "\n")
10 cat("standard variance: ", std_deviation, "\n")
11 cat("skewness: ", skewness_value, "\n")
12 cat("kurtosis: ", kurtosis_value, "\n")
```

运算结果:

```
1 mean: 158.8148
2 variance: 38.74516
3 standard variance: 6.224561
4 skewness: -1.587447
5 kurtosis: 5.212256
```

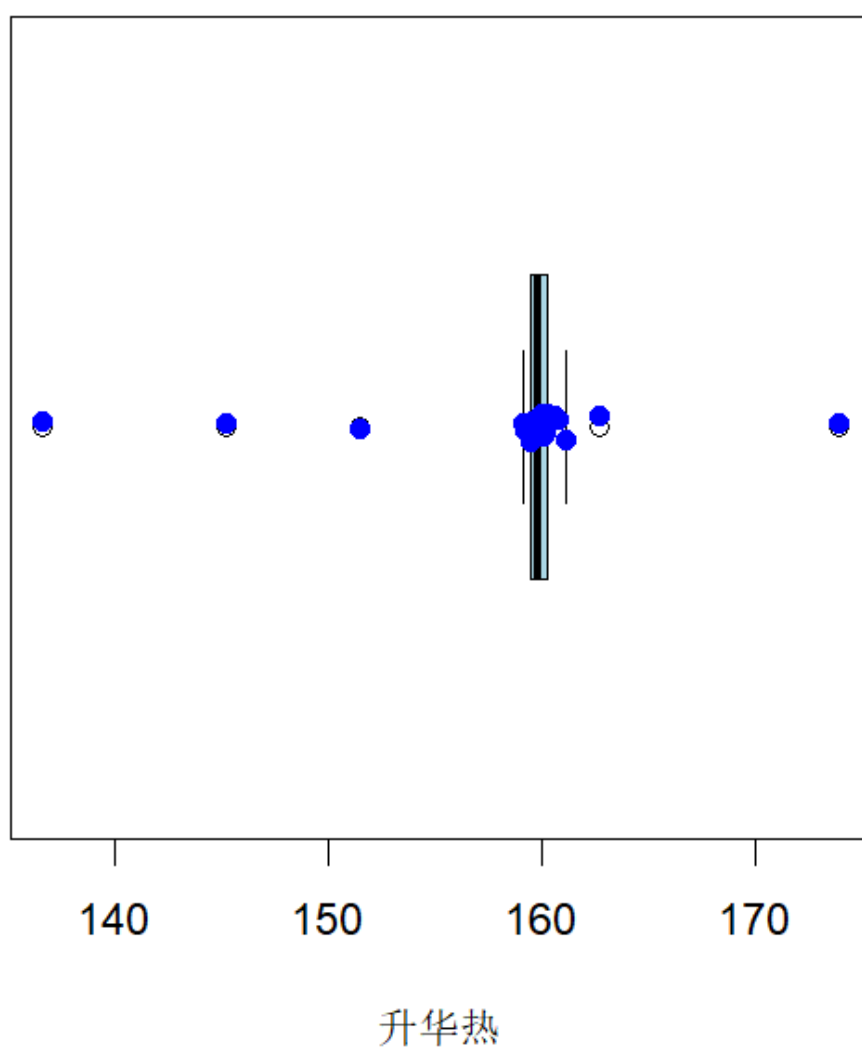
(5) 制作箱线图

R 代码:

```
1 # 使用boxplot()函数创建箱线图
2 boxplot(sublimation_heat_data,
3         main="金融铱升华热的箱线图",
4         xlab="升华热",
5         col="lightblue", # 设置箱体颜色
6         notch=FALSE, # 如果为TRUE, 则在箱体中添加一个缺口以表示中位数的置信区间
7         horizontal=TRUE # 设置箱线图的方向, FALSE为垂直, TRUE为水平
8     )
9
10 # 添加数据点, 以更清楚地展示所有观测值
11 points(sublimation_heat_data,
12        jitter(rep(1, length(sublimation_heat_data))),
13        col="blue", pch=16)
```

运行结果:

金融铱升华热的箱线图



THE END